

Directory of 21 Pre-Indexed Indian Standards (IS)

Bureau of Indian Standards Knowledge Base • Smart India Hackathon 2026 (SIH26107)

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1. Summary Table of All 21 Standards

#	Standard IS Number	Sector / Category	Specification Title	QCO Mandate
1	IS 10500:2012	Food & Water Safety	Drinking Water Specification	Mandatory (QCO)
2	IS 12254:2021	Personal Protective Equipment	Polyyvinyl Chloride (PVC)	Mandatory (QCO)
3	IS 1293:2019	Electrical & Electronics	Plugs and Sockets Outlets	Mandatory (QCO)
4	IS 14543:2024	Food & Water Safety	Packaged Drinking Water	Mandatory (QCO)
5	IS 15298 (Part 2):2016	Personal Protective Equipment	Personal Protective Equipment	Mandatory (QCO)
6	IS 15477:2019	Construction & Infrastructure	Adhesives for Use with	Voluntary
7	IS 15644:2006	Toys & Children Products	Safety of Electrical Toys	Mandatory (QCO)
8	IS 15652:2006	Electrical & Electronics	Insulating Mats for El	Mandatory (QCO)
9	IS 15700:2018	Management Systems & Services	Quality Management Sys	Voluntary
10	IS 15885 (Part 2/Sec 13):2012	Electrical & Electronics	Safety of Lamp Con	Mandatory (QCO)
11	IS 16018:2021	Automotive & Road Safety	Helmets for Riders of	Mandatory (QCO)
12	IS 16240:2023	Food & Water Safety	Reverse Osmosis (RO) B	Mandatory (QCO)
13	IS 1786:2008	Construction & Infrastructure	High Strength Deformed	Mandatory (QCO)
14	IS 2347:2017	Kitchen & Home Appliances	Domestic Pressure Cook	Mandatory (QCO)
15	IS 269:2015	Construction & Infrastructure	Ordinary Portland Ceme	Mandatory (QCO)
16	IS 302 (Part 1):2008	Electrical & Electronics	Safety of Household an	Mandatory (QCO)
17	IS 3196 (Part 1):2013	Cylinders & Pressure Vessels	Welded Low Carbon Stee	Mandatory (QCO)
18	IS 4985:2021	Pipes & Fittings	Unplasticized PVC Pipe	Mandatory (QCO)
19	IS 694:2010	Electrical & Electronics	PVC Insulated Cables f	Mandatory (QCO)
20	IS 7318 (Part 1):2020	Industrial & Manufacturing	Approval Testing of We	Voluntary
21	IS 9873 (Part 1):2019	Toys & Children Products	Safety of Toys Part	Mandatory (QCO)

2. Detailed Technical Clauses & Testing Tolerances

1. IS 10500:2012 — Drinking Water — Specification

Category: Food & Water Safety | **Scheme:** National Benchmark Standard | **QCO:** **Mandatory (QCO)**

Scope: Prescribes the acceptable and permissible requirements and limits for essential characteristics and desirable parameters of drinking water supplied through municipal piped networks and rural water supply schemes.

- **pH Value:** Acceptable: 6.5 to 8.5
- **Total Dissolved Solids (TDS):** Acceptable: 500 mg/l max (Permissible: 2000 mg/l)
- **Fluoride and Nitrate:** Fluoride: 1.0 mg/l max; Nitrate: 45 mg/l max
- **Consumer Tip:** If you tap water TDS is below 300 mg/L with normal pH, simple UV/UF filtration is adequate without heavy RO water waste.

2. IS 12254:2021 — Polyvinyl Chloride (PVC) — Industrial Boots — Specification

Category: Personal Protective Equipment | **Scheme:** Scheme-I (Mandatory ISI Mark Certification) | **QCO:** **Mandatory (QCO)**

Scope: Covers requirements for water-proof moulded PVC industrial boots intended for protection against water, acids, alkalis, oil, and mud in mines and chemical plants.

- **Flexing Resistance:** Zero crack growth > 4 mm
- **Chemical Resistance:** Tensile strength retention > 80%
- **Consumer Tip:** Ensure industrial gumboots have the authentic ISI stamp on the upper rim.

3. IS 1293:2019 — Plugs and Socket-Outlets of Rated Voltage up to 250 V and Current up to 16 A — Specification

Category: Electrical & Electronics | **Scheme:** Scheme-I (Mandatory ISI Mark Certification) | **QCO:** **Mandatory (QCO)**

Scope: Covers plugs and fixed or portable socket-outlets for AC only, rated from 2.5 A up to 16 A / 250 V for household and commercial electrical installations.

- **Safety Shutter Engagement Test:** 1 mm probe must not touch live parts under 20 N force
- **Temperature Rise Test:** Terminal temp rise ≤ 45 K
- **Consumer Tip:** Never use ungrounded 2-pin plugs on metal-cased appliances; use ISI 3-pin plugs.

4. IS 14543:2024 — Packaged Drinking Water (Other than Packaged Natural Mineral Water) — Specification

Category: Food & Water Safety | **Scheme:** Scheme-I (Mandatory ISI Mark & In-house Laboratory) | **QCO:** **Mandatory (QCO)**

Scope: Prescribes the requirements and methods of sampling and testing for packaged drinking water filled in hermetically sealed food-grade containers of various capacities (pouches, bottles, jars, cans).

- Microbiological Sterility Testing:** Zero (Absent) per 250 ml sample for all pathogen classes
- Total Dissolved Solids (TDS):** 75 mg/l to 500 mg/l
- Pesticide Residues (Individual & Total):** Individual pesticide < 0.0001 mg/l (0.1 ppb); Total < 0.0005 mg/l (0.5 ppb)
- Consumer Tip:** Always check for the holographic/tamper-evident seal and valid ISI mark before buying water bottles.

5. IS 15298 (Part 2):2016 — Personal Protective Equipment — Part 2: Safety Footwear

Category: Personal Protective Equipment | **Scheme:** Scheme-I (Mandatory ISI Mark) | **QCO:** **Mandatory (QCO)**

Scope: Specifies basic and additional requirements for safety footwear used for commercial and industrial workplaces, equipped with safety toe caps.

- Toe Cap Impact Resistance:** Clearance beneath toe cap ≥ 14 mm
- Penetration Resistance of Sole:** Spike must not penetrate insole barrier
- Consumer Tip:** S1P or S3 safety shoes include both 200J steel toe cap and steel anti-nail penetration midsole.

6. IS 15477:2019 — Adhesives for Use with Ceramic, Mosaic and Stone Tiles — Specification

Category: Construction & Infrastructure | **Scheme:** Scheme-I (ISI Mark Certification) | **QCO:** **Voluntary**

Scope: Covers polymer-modified cementitious adhesives for fixing ceramic, vitrified, mosaic, and natural stone tiles on interior and exterior walls and floors.

- Tensile Adhesion Strength:** Type 2: ≥ 1.0 N/mm²; Type 4: ≥ 1.5 N/mm²
- Slip Resistance:** Slip ≤ 0.5 mm
- Consumer Tip:** For large vitrified tiles or wall cladding, regular cement slurry fails; use Type 2 or Type 3 ISI tile adhesive.

7. IS 15644:2006 — Safety of Electric Toys

Category: Toys & Children Products | **Scheme:** Scheme-I (ISI Mark Certification) | **QCO:** **Mandatory (QCO)**

Scope: Covers the safety requirements for electric toys operating from batteries, transformers, or solar cells with rated supply voltage not exceeding 24 V DC/AC.

- Operating Voltage Limit Test:** Nominal voltage ≤ 24 V
- Thermal Heating & Battery Compartment Test:** Battery surface temp rise < 45 K; Casing < 25 K
- Consumer Tip:** Never mix old and new batteries in electric toys.

8. IS 15652:2006 — Insulating Mats for Electrical Purposes — Specification

Category: Electrical & Electronics | **Scheme:** Scheme-I (Mandatory ISI Mark) | **QCO:** **Mandatory (QCO)**

Scope: Covers elastomer/polymer insulating floor mats used near high-voltage electrical switchboards for protection against electric shock up to 33 kV.

- Dielectric Strength & Proof Voltage:** Class A — (3–3 kV) — 10 kV; Class B — (1–1 kV) — 22 kV; Class C — (3–3 kV) — 36 kV
- Leakage Current Test:** Leakage < 10 μ A/m²
- Consumer Tip:** Obsolete rubber mats under IS 5424 are banned; use synthetic elastomer mats with IS 15652.

9. IS 15700:2018 — Quality Management Systems — Requirements for Service Quality by Public Service Organizations (Sevottam) —

Category: Management Systems & Services | **Scheme:** Management Systems Certification Scheme (MSCS) | **QCO:** **Voluntary**

Scope: Specifies requirements for a quality management system where a public service organization needs to demonstrate citizen-centric service delivery.

- Citizen Charter Implementation Audit:** 100% published charter commitments with KPIs
- Public Grievance Redressal Audit:** Time-bound resolution within 30 days
- Consumer Tip:** Sevottam charters bind government offices to deliver citizen services within defined days.

10. IS 15885 (Part 2/Sec 13):2012 — Safety of Lamp Control Gear — Electronic Control Gear for LED Modules — (LED Drivers) —

Category: Electrical & Electronics | **Scheme:** CRS (Compulsory Registration Scheme - R-XXXXXXX) | **QCO:** **Mandatory (QCO)**

Scope: Covers safety requirements for electronic control gear for DC supplies up to 250 V or AC supplies up to 1000 V for constant-voltage or constant-current LED modules.

- Surge Protection Immunity Test:** Withstand min 2.5 kV (indoor) or 4 kV (outdoor)
- Total Harmonic Distortion (THD):** THD $\leq$ 10% to 25%
- Consumer Tip:** Check for R-XXXXXX registration mark on LED power supplies to prevent surge fire hazards.

11. IS 16018:2021 — Helmets for Riders of Two-Wheeled Motor Vehicles — Specification

Category: Automotive & Road Safety | **Scheme:** Scheme-I (Mandatory ISI Mark) | **QCO:** **Mandatory (QCO)**

Scope: Prescribes requirements for protective helmets for everyday use by drivers and passengers of two-wheeled motor vehicles to safeguard against head injuries.

- Impact Attenuation Test:** Peak head acceleration $\leq$ 275 g
- Retention System Dynamic Test:** Dynamic extension $\leq$ 35 mm; Residual $\leq$ 25 mm
- Consumer Tip:** Wearing a non-ISI helmet is illegal in India under the Motor Vehicles Act.

12. IS 16240:2023 — Reverse-Osmosis-(RO)-Based-Point-of-Use-Water-Treatment-Systems — Specification

Category: Food & Water Safety | **Scheme:** Scheme-I (Mandatory ISI Mark Certification) | **QCO:** **Mandatory (QCO)**

Scope: Prescribes requirements for residential point-of-use RO water purifiers, mandating minimum water recovery rates, TDS reduction efficiency, and total microbiological safety.

- **Water Recovery Efficiency Test:** Minimum 40% recovery
- **TDS Reduction Percentage:** Minimum 85% TDS reduction
- **Consumer Tip:** Look for IS-16240 ISI mark to ensure your RO purifier does not excessively waste water.

13. IS 1786:2008 — High-Strength-Deformed-Steel-Bars-and-Wires-for-Concrete-Reinforcement — Specification-(TMT-Rebars)

Category: Construction & Infrastructure | **Scheme:** Scheme-I (Mandatory ISI Mark Certification) | **QCO:** **Mandatory (QCO)**

Scope: Covers the requirements of deformed steel bars and wires for use as reinforcement in concrete in grades Fe-415, Fe-415D, Fe-500, Fe-500D, Fe-550, Fe-550D produced via Thermomechanical Treatment-(TMT).

- **0.2% Proof Stress / Yield Stress:** Fe-500D: Min 500.0 N/mm²; Fe-550D: Min 550.0 N/mm²
- **Tensile Strength to Yield Strength Ratio -(TS/YS):** Fe-500D: TS/YS ≥ 1.10; Fe-550D: TS/YS ≥ 1.08
- **Percentage Elongation at Fracture:** Fe-500D: Min 16.0%; Fe-550D: Min 14.5%
- **Bend and Re-bend Test:** No transverse crack or rupture visible on outer curved surface
- **Consumer Tip:** For home construction in seismic zones, always insist on Fe-500D or Fe-550D grade TMT bars with D (Deformed) symbol.

14. IS 2347:2017 — Domestic-Pressure-Cookers — Specification

Category: Kitchen & Home Appliances | **Scheme:** Scheme-I (ISI Mark Certification) | **QCO:** **Mandatory (QCO)**

Scope: Specifies requirements for domestic pressure cookers made of aluminium alloy or stainless steel with capacities from 1L to 15L, operating pressures 0.5 to 1.1 bar.

- **Operating Pressure Test:** Within ±10% of declared operating pressure (typically 1.0 kg/cm²)
- **Proof Pressure Test:** Must withstand 2.5 times operating pressure without permanent distortion.
- **Bursting Pressure Test:** Shall not burst at less than 4.5 times normal working pressure.
- **Safety Valve Operating Test:** Releases pressure between 1.5 to 2.2 times operating pressure.
- **Consumer Tip:** Always verify the 7 or 8 digit CM/L licence number below the ISI logo using the BIS Care App.

15. IS 269:2015 — Ordinary Portland Cement — (33, — 43 — and 53 — Grade) — Specification

Category: Construction & Infrastructure | **Scheme:** Scheme-I (Mandatory ISI Mark Certification) | **QCO:** **Mandatory (QCO)**

Scope: Covers the manufacture, chemical and physical requirements of ordinary Portland cement in 33 — Grade, 43 — Grade, and 53 — Grade.

- **Compressive Strength — (28 — Days):** Grade 53: Min 53.0 MPa at 28 days; Grade 43: Min 43.0 MPa
- **Initial and Final Setting Time:** Initial setting time $\geq$ 30 min; Final setting time $\leq$ 600 min
- **Consumer Tip:** Never use cement bags older than 90 days from packing date without re-testing strength.

16. IS 302 (Part 1):2008 — Safety of House-hold and Simi-lar — Electrical Appliances — General — Requirements

Category: Electrical & Electronics | **Scheme:** Scheme-I (ISI Mark) / CRS | **QCO:** **Mandatory (QCO)**

Scope: Deals with safety of electrical appliances for house-hold purposes, rated voltage not more than 250 V single phase (electric irons, immersion heaters, mixer-grinders, toasters, geysers).

- **High Voltage Electric Strength Test:** Withstand 1250 V AC (Class I) or 3750 V AC (Class II) for 1 minute
- **Leakage Current & Earth Continuity:** Earth resistance $< 0.1 \Omega$; Leakage current < 0.75 mA for Class I portable appliances
- **Thermal Heating & Abnormal Operation:** Enclosure temp rise must not exceed safe limit (60 K for handles)
- **Consumer Tip:** Check for 3-pin earthed plugs on high-wattage heating appliances (irons, geysers, room heaters).

17. IS 3196 (Part 1):2013 — Welded Low Carbon Steel — Cylinders for Low Pressure — Liquefiable — Gases — Part 1: Cylinders for LPG

Category: Cylinders & Pressure Vessels | **Scheme:** Scheme-I (PESO & ISI Joint Statutory Licensing) | **QCO:** **Mandatory (QCO)**

Scope: Specifies requirements for design, materials, manufacture, process, heat treatment, testing and inspection of refillable welded low carbon steel cylinders of capacity up to 150 litres water capacity for liquefied petroleum gases (LPG).

- **Hydrostatic Stretch Test:** Permanent expansion must not exceed 1.0% of total expansion at 2.5 MPa (25 bar)
- **Hydrostatic Burst Pressure Test:** Must not burst below 4.5 MPa (45 bar) and failure must occur longitudinally without fragmentation
- **Consumer Tip:** Check the three stay plates on your LPG cylinder for the clear ISI mark and PESO certification stamp.

18. IS 4985:2021 — Unplasticized PVC Pipes for Potable Water Supplies — Specification**Category:** Pipes & Fittings | **Scheme:** Scheme-I (ISI Mark Certification) | **QCO:** **Mandatory (QCO)****Scope:** Prescribes requirements for unplasticized polyvinyl chloride (UPVC) pipes for potable water supplies in domestic, agricultural, and municipal plumbing.

- Hydrostatic Pressure Test:** No burst or leak at 4.2 times rated pressure for 1 hour.
- Lead & Toxic Metal Extraction:** Lead extraction < 0.05 mg/l on third extraction.
- Consumer Tip:** Ensure pipes for drinking water are lead-free UPVC certified under IS 4985.

19. IS 694:2010 — PVC Insulated Cables for Working Voltages up to and including 1100 V — Specification**Category:** Electrical & Electronics | **Scheme:** Scheme-I (ISI Mark Certification) | **QCO:** **Mandatory (QCO)****Scope:** Covers the requirements for single-core and multi-core PVC insulated and sheathed/unsheathed electrical cables with copper or aluminium conductors for working voltages up to 1100 V AC, widely used in domestic building wiring and industrial panels.

- Conductor Resistance Test:** Max resistance e.g. 12.1 Ω /km for 1.5 sq mm class 2 copper conductor.
- Flame Retardance Test (Oxygen & Temperature Index):** Oxygen index > 29% for FR/FR-LS cables; Temperature index > 250°C.
- Consumer Tip:** Never compromise on domestic electrical wiring; counterfeit cables with sub-standard copper cause fires.

20. IS 7318 (Part 1):2020 — Approval Testing of Welders for Fusion Welding — Part 1: Steels**Category:** Industrial & Manufacturing | **Scheme:** Personnel & Process Certification | **QCO:** **Voluntary****Scope:** Prescribes requirements for welder qualification and approval testing in fusion welding of carbon and alloy steels across all standard welding positions (1G to 6G).

- Visual Examination of Weld Seam:** Zero crack, zero lack of fusion, undercut $\leq$ 0.5 mm.
- Radio-graphic Examination (NDT):** Acceptance level B under ISO 5817.
- Consumer Tip:** For critical building/bridge fabrication, ensure welders hold active IS 7318 certification.

21. IS 9873 (Part 1):2019 — Safety of Toys — Part 1: Safety Aspects Related to Mechanical and Physical Properties

Category: Toys & Children Products | **Scheme:** Scheme-I (Mandatory Factory Audit & ISI Mark) | **QCO:** **Mandatory (QCO)**

Scope: Applies to all toys intended for use by children under 14 years of age. Specifies criteria for physical features including sharp edges, points, small parts (choking hazard for under 36 months), cords, and impact resistance.

- Small Parts Cylinder Choking Test:** No detachable small parts for toys intended for children under 3 years.
- Sharp Edge & Sharp Point Test:** Zero hazardous sharp edges or sharp points.
- Drop and Impact Test:** Must not crack, shatter into sharp splinters, or release small parts.

Consumer Tip: Every toy sold in Indian retail or online must have the ISI mark; importing or selling non-ISI toys is illegal in India.